

EVALUATION OF MOORING LINE DEGRADATION IN FLOATING OFFSHORE WIND TURBINES: A COMPREHENSIVE ANALYSIS OF CORROSION AND FATIGUE UNDER ENVIRONMENTAL IMPACTS

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ABSTRACT

As the demand for renewable energy grows, floating offshore wind turbines have become crucial, especially in deep water areas where traditional structures are impractical. These turbines need station keeping systems, which often consist in catenary mooring lines, making their durability vital against extreme weather conditions. While direct monitoring of these lines would be preferable, it is often impractical due to high costs and maintenance. This research presents a new method to evaluate the degradation of mooring lines. The study investigates fatigue and corrosion behavior along the mooring lines by simulating environmental impacts, considering varied forces and weather patterns. Considering seawater's oxygen and temperature, a distinct corrosion model is merged with fatigue performance predictions based on DNV standards. Mooring line loading conditions derived from environmental factors are then incorporated into a finite element model, assessing various load cases and corrosion's role in line degradation. This comprehensive model examines the combined effects of corrosion (in particular, pitting corrosion) and fatigue, including hydrostatic pressure and out-of-plane bending on the performance of the mooring line chain in the offshore environment. The methodology's validation relies on previous research and mooring line operational data.

Keywords: Offshore Wind Turbine, Mooring line, Fatigue, Corrosion, Structural Analysis, Extreme loads

NOMENCLATURE

a	Intercept parameter
C_R	Corrosion Rate
C_D	Corrosion Depth
d	Diameter
d_0	Reference heigh

L_i	Amplitude of load
m	S-N curve slope
n_i	Cycles of Load
N	Lifecycles
N_{ref}	Reference Cycle Number
P	Hydro static pressure
R	Load Ratio
t	Immersion time
$V_{c,tide}$	Current velocity at ref. depth
$V_{c,wind}$	Wind speed at ref. depth
z	Depth
$\Delta\sigma$	Stress range
σ_{max}	Maximum Stress
ρ	Density of Liquid

1. INTRODUCTION

Floating offshore wind turbines (FOWT) are emerging in the renewable energy sector, enabling energy to be harvested in deep water [1]. Unlike conventional offshore wind turbines, FOWT are not installed on a monopile on the seabed, but on a substructure like floating pontoons, held in place by mooring lines, and countries characterized by limited shelf seas could benefit from this solution [2].

The first floating offshore wind farm was installed in 2017, and since then, the installed capacity of floating offshore wind in Europe has reached 73.33 MW [3]. With innovation, often new problems arise. As reported by Fontaine et al. [4], half of the turbine failures were caused by mooring line failure, of which half were caused by corrosion and fatigue. These numbers reflect the situation in the oil and gas industry, which has been using mooring lines to secure floating production units since 1977 [5]. This problem is not new to the offshore industry but has been an

obstacle for years and it has resulted in lost investment for decades [4].

Extensive research has been conducted to enhance the understanding of this issue, leading to the development of advanced mooring line models. These models are crucial for pinpointing the fundamental causes and potential solutions for this problem. Corrosion significantly diminishes the service life of mooring lines submerged in seawater. This corrosion is primarily attributed to several environmental factors, including the presence of oxygen, variations in temperature, bacterial activity, water salinity, and seawater's general impact on steel [6].

Corrosion is found to reduce the lifetime of the mooring line through the weakening of the material. In the development of the corrosion in steel, corrosion pits are created first, followed by the uniform corrosion [7]. Uniform corrosion uniformly impacts the entire metal surface, resulting in a gradual and predictable decrease in diameter of the mooring line. As described by Du et al. [8] the diameter reduction is calculated as 0.4 mm/year, as 0.5 mm/year in combination with wear [9], or even up to 0.6 mm/year in [10]. In contrast, pitting corrosion creates localized, deep penetrations in the metal surface, leading to severe damage at specific points on the structure [6]. For analysis of pitting corrosion, the condition of the mooring line is assessed on a scale of 1-7. One is the best condition without corrosion and seven is the most critical condition with pitting corrosion. This localized nature of pitting corrosion poses a greater threat to structural integrity than the more evenly distributed effects of uniform corrosion.

Research on mooring line corrosion is often restricted by the substantial resources needed to analyze the entire length of a mooring line. Consequently, studies are typically confined to examining the line at selected points rather than conducting a comprehensive assessment. An often-neglected impact is how corrosion pits develop along the mooring line. As described by Orlikowski et al. [11], the factors influencing corrosion vary with water depth, which could lead to a variation in the corrosion of the mooring line alone depending on water depth.

Fatigue is often caused by cyclic loading, leading to crack initiation and development. These cracks may form either at corrosion pits or at critical high stress sections of the mooring line and then develop through cyclic loading [8]. They can significantly reduce the service life of a mooring line. If the cracks develop to a critical depth, the mooring line may tear, leading to dangerous failure events [12].

A combination of fatigue and corrosion can reduce service life and must be considered. Stress-Life (S-N) curves are used to determine the fatigue life of a material [13]. They describe how many cycles in a stress range a material can withstand before fatigue occurs. As presented by Lone et al. [14], new S-N curves have been developed to include corrosion in the fatigue life assessment. A notable issue with these predictive models is the conservative treatment of corrosion as a constant factor [15].

Treating corrosion as a constant factor may lead to a significant underestimation of its true impact, especially considering the variability in water depth and immersion time. This simplistic approach overlooks the interplay of environmental factors. Furthermore, research into how corrosion and fatigue interact to create specific weak points along the mooring line is notably limited. This oversight raises an important question: does the variability in corrosion and fatigue intensity along the mooring line contribute to more pronounced deterioration in certain segments? While some studies have identified specific regions of the mooring line as being more failure-prone, a detailed investigation into the combined effects of corrosion and fatigue in these areas remains unexplored. This gap in research underscores the need for more comprehensive studies to understand the localized effects of these mechanisms and develop more effective strategies for mooring line maintenance and design.

Consequently, our study emphasizes the development of predictive Finite Element (FE) models to assess the impact of corrosion pits on the fatigue life of mooring lines operating in deep-water environments. A method is proposed how to analyse the combined effects of corrosion pits and fatigue along FOWT mooring lines. The analysis is carried out using a corrosion model based on environmental factors varying with water depth and a load analysis carried out using an open-source software-engineering framework for FAST-based aero-hydro-servo-elastic engineering models of wind-turbines and wind-plants (OpenFAST¹) and FE method. Additionally, the weak points along the mooring lines with the higher failure probability are determined. The main areas of concern subjected to a intensive dynamic loading and corrosion pits are analysed.

To determine all the chain specifications and details and the environmental data, the COREWIND project was selected, and the results were compared with other literature studies.

The paper is organized as follows. Section 2 presents the case study on which the research is based. Section 3 describes the proposed methodology, Section 4 presents and discusses the results, and lastly Section 5 gives a conclusion of the paper.

2. CASE STUDY

The effectiveness of the proposed methodology is demonstrated by utilizing the COREWIND project, which is funded by the European Union and offers open-source FOWT data, as reported in [16]. The project involves the design of a 15 MW wind turbine tailored to various substructures and locations, aiming to explore potential cost reductions.

Specifically, the degradation analysis focuses on this 15 MW turbine paired with the Activefloat substructure, situated southeast of Gran Canaria. Details about the wind turbine's configuration are illustrated in Figure 1.

The full specifications for the 15 MW wind turbine are defined in [15]. For completeness, we report the main features relevant to this study. It is engineered for energy production at a

¹ OpenFAST v2.6, <https://github.com/openfast/openfast/>, accessed: May 4, 2023 (2021).

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cut-in wind speed of 4 m/s and achieves its rated capacity at a wind speed of 10.77 m/s, capable of withstanding up to a cut-off wind speed of 25 m/s. This turbine features a direct drive generator and is designed with a substantial rotor radius of 120 meters and a hub height of 150 meters. This turbine is supported by the innovative Activefloat substructure, a pontoon-like framework with three columns floating from 26 m below sea level to 9 m above it, as shown in Figure 2. The substructure is anchored at a depth of 15 m by a mooring system. This system is arranged at a 120-degree angle and comprises three catenary mooring lines, each wholly constructed from 160 mm diameter R4 steel chain segments. The precise measurements of these mooring chain segments are depicted in Figure 3. The mooring material R4 is high-strength steel with the properties [16] presented in Table 1.

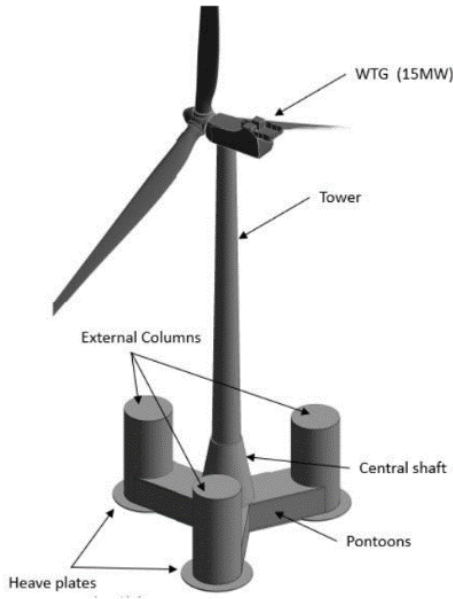


FIGURE 1: COREWIND 15MW TURBINE ON SEMI-SUBMERSIBLE STRUCTURE USED AS A CASE STUDY

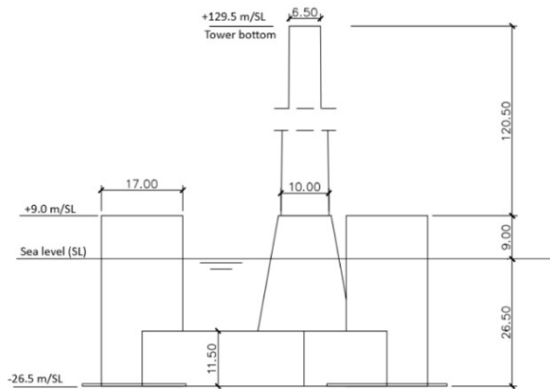


FIGURE 2: SIDE VIEW OF THE SEMI-SUBMERSIBLE FLOATER IN THE DIMENTIONS OF METERS [13]

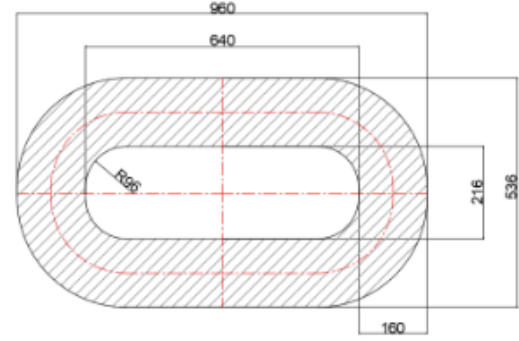


FIGURE 3: MOORING CHAIN DIMENSIONS IN MILLIMETERS

Considering the specified dimensions and the material properties of the mooring chain (Table 1), the Minimum Breaking Load (MBL) is determined to be 17 MN. At the point where the mooring lines meet the seabed, they are securely connected with embedded anchors, which are designed to provide reliable holding capacity in the marine environment.

2.1 Environmental Data

In the considered case study, the load cases from the COREWIND project have been selectively filtered to focus on the most prevalent scenarios that affect mooring line degradation. A comprehensive list of these load cases is available in [15]. Table 2 provides a detailed breakdown of the wind data, categorizing the information by speed and direction. Similarly, Table 3 presents the wave data, organized by height, peak period, and direction, offering a clear depiction of the marine conditions relevant to the study.

TABLE 1: MATERIAL PROPERTIES OF R4 MOORING STEEL

Material Name	R4
Mass Density [Te/m^3]	7.98
Young's modulus [GPa]	210
Poisson ratio	0.3

TABLE 2: ENVIRONMENTAL WIND DATA

	Range		Probability [%]
Wind speed range [m/s]	6.0	7.5	21.35
Wind direction [radians]	25		45

TABLE 3: ENVIRONMENTAL WAVE DATA

	Range		Probability [%]
Wave height [m]	1	1.5	21.35
Wave peak period [s]	6	7	
Wave direction [radians]	1		45

The current is analyzed as two distinct components: tidal and wind-induced. For this study, the tidal current is assumed to have a direction of 202 or 22 degrees. The direction of the wind-induced current is determined by the data provided in Table 2. The flow velocities for both the wind and tidal currents are quantified and detailed in Equations 1 and 2.

$$v_{c,wind} = v_{c,wind}(0) \left(\frac{d_0 + z}{d_0} \right) \text{ for } -d_0 \leq z \leq 0 \quad (1)$$

$$v_{c,tide} = v_{c,tide}(0) \left(\frac{d_0 + z}{d_0} \right)^{\frac{1}{7}} \text{ for } z \leq 0 \quad (2)$$

For the assessment of corrosion-related degradation, data on oxygen concentration, salinity, and temperature were sourced from the World Ocean Atlas (WOA) [17]. The extracted data corresponds to a data point collected in February 2020 at a depth of 200 meters, situated merely 14 kilometers from the COREWIND project's site, thus providing a reliable representation of the local conditions. Figure 4 visualizes the variation of oxygen levels and temperature from this data set, illustrating a clear trend of decreasing oxygen concentration and temperature with increasing depth.

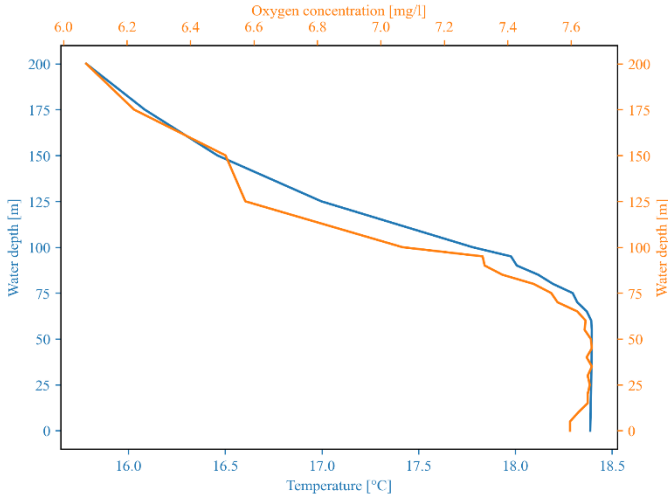


FIGURE 4: OXYGEN AND TEMPERATURE DATA AT THE COREWIND LOCATION

3. METHODOLOGY

This study employs a combination of the FE method and hot spot analysis by DNVGL-RP-C203 standard [18] to evaluate the fatigue and pitting corrosion resistance along the mooring line. The analysis started with identifying load cases (Section 3.1), utilizing environmental data and the multi-fidelity simulation tool, open-source wind turbine simulation tool OpenFAST [19] to generate the load history along the mooring line. This load history represents the endured loads on a specific mooring line area over a simulation tool. Subsequently, the load history was converted into Damage Equivalent Loads (DEL). DEL's represent the combined impact of the loads in the load history on the mooring line section. These were then used for FE analysis in computer-aided engineering software (ABAQUS) to the designated areas of the mooring line model. In parallel, a corrosion model utilizing corrosion pits was developed (Section 3.2), informed by water temperature and oxygen concentration data. This model was integrated with a long-term immersion corrosion model to account for the effects of sustained exposure to seawater. The modeled corrosion pitting was incorporated into

the mooring line section FE model (Section 3.3.) for calculating stress within the mooring chain, providing a comprehensive understanding of the degradation phenomena.

3.1 Load Cases Description

Load implementation within the FE analysis was carried out by analyzing the mooring section, with calculations performed using OpenFAST to determine mooring line loads based on various environmental factors that influence FOWT. We implemented a model based on the IEA 15MW wind turbine in the OpenFast software to calculate mooring line loads. This open-source IEA 15MW model has been used in prior projects, but we can confirm its validity by comparing it to the COREWIND project, particularly [10], which also calculated the DELs for various mooring line sections.

From this, we can draw two conclusions. The range of the DELs has an acceptable magnitude, like that determined by the authors of [20]. Secondly, equivalent loads at the anchor section can surpass those at the fairlead. This indicates that the change in damage equivalent load along the mooring line is likely.

The computational process within OpenFAST involves a suite of software subtools supplied by the COREWIND project, which are tailored for the simulation of wind turbines and their substructures. Modules such as AERODYN for aerodynamic analysis and SERVODYN for control systems were utilized in their published form. In contrast, the MOORDYN module, designated for mooring analysis, was customized by the authors to incorporate a chain representation with an equivalent diameter of 288mm. This adjustment was necessary to simulate the mooring line as a steel rod accurately for the load calculations. Additionally, within the MOORDYN model, specific nodes were established to serve as points where the mooring line forces could be outputted. At these nodes, a time history analysis was conducted to determine a DEL for each. This was achieved by employing the Rainflow counting method and applying the Miner's rule to assess fatigue damage [21]. These steps allow for a detailed quantification of the cumulative damage over time, providing vital insights into the longevity and maintenance requirements of the mooring system.

To accurately assess the load history, the Rainflow counting method is applied to enumerate the amplitudes and cycles of stress the mooring line endures. Building upon this, the Miner equation takes the results of the Rainflow count—specifically the load cycles and their respective amplitudes—and computes the total accumulated damage. It factors in the cumulative effect of all load cycles and their intensities. The comprehensive formula that integrates the Rainflow count and Miner's damage accumulation is articulated in Equation 3.

$$S_{DEL} = \sqrt[m]{\frac{1}{N_{ref}} \sum n_i S_i^m} \quad (3)$$

3.2 Pitting Corrosion and Fatigue Model

To determine the corrosion development along the mooring line the authors utilized a corrosion model based on the model by Nevshupa et al. [6]. This model leverages the influence of oxygen concentration and temperature on corrosion kinetics to establish a rate of corrosion in millimeters per year, specifically for R4 grade mooring steel in synthetic seawater. Therefore, the data in [6] is based on the corrosion progression which is uniformly distributed, the present study utilizes the corrosion rate to determine the depth of the corrosion pit. To ensure the relevance of the model to the COREWIND project's location, data proximate to the site were obtained from the WOA. The 2019 measurements at a depth of 200 meters provided a precise representation of the oxygen and temperature conditions. These parameters were integrated into the Nevshupa et al. model, allowing the authors to calculate a distinct corrosion rate at various water depths. The resulting corrosion rates are systematically presented in Figure 5.

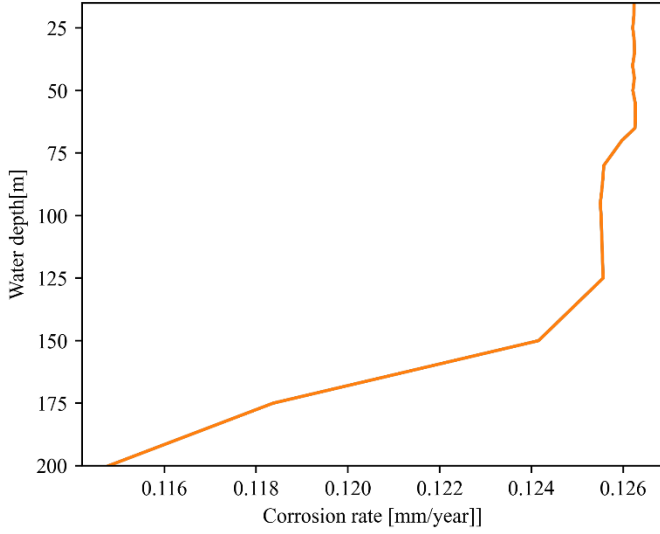


FIGURE 5: CORROSION RATE IN REGARD TO WATER DEPTH [6].

As this corrosion rate is mainly based on short term immersion, to extend its applicability to long-term scenarios, it was decided to integrate this rate with a long-term immersion model. After evaluating various methodologies [21], the power law formula proposed by Melchers et al. was selected, as described in Equation 4:

$$C_D = C_R t^\alpha \quad (4)$$

In Equation 4, the corrosion depth (C_D) is calculated by multiplying the corrosion rate (C_R) by the immersion time (t), raised to a chosen exponent (α). For comparison, the long-term immersion model was examined under three different scenarios: using the Fickian diffusion model ($\alpha = 0.5$), a linear approach ($\alpha = 1$), and a custom exponent chosen by the authors ($\alpha = 0.8$). These varying models are depicted in Figure 6. After thorough consideration, the authors proceeded with the Fickian diffusion model [21] for further analysis.

In the analysis of the FE results, which included both corrosion and load data, the derived stresses were plotted on S-N curves based on Equation 5

$$\log(N) = \log(a) - m \log(\Delta\sigma) \quad (5)$$

where

$$\log(\Delta\sigma) = \sigma_{max} - R\sigma_{max} \quad (6)$$

and the logarithm of the lifecycles (N) is calculated as the logarithm of the intercept parameter (a) of the curve, subtracted by the product of the slope parameter (m) and the logarithm of the stress range ($\Delta\sigma$). The stress range is defined by the maximum stress (σ_{max}) adjusted by the load ratio R .

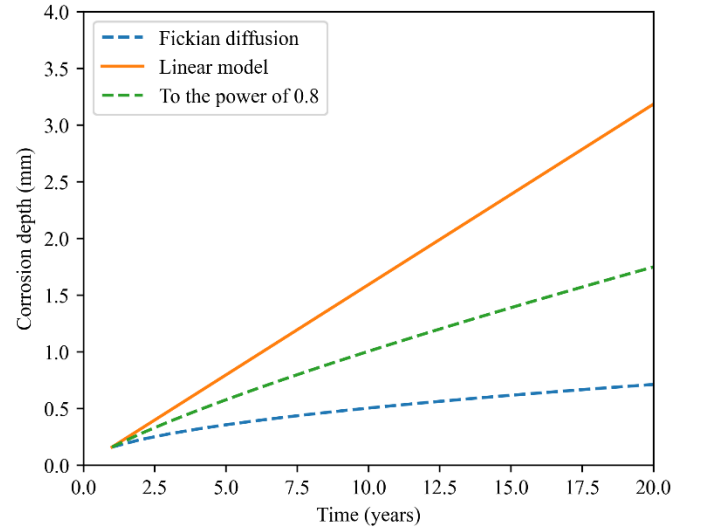


FIGURE 6: CORROSION DEPTH BASED ON VARIATIONS IN THE POWER LAW

To provide a comprehensive analysis, three different S-N curves, each corresponding to distinct slope parameters, were evaluated. These curves are delineated in standards and were compared, as illustrated in Figure 7. The DNV standards include scenarios in air, seawater with cathodic protection, and free corrosion conditions.

After conducting a comparative analysis of the three standard S-N curves, it is possible to observe that the curve corresponding to the free corrosion scenario yielded the lowest life cycle expectancy.

Therefore, in Section 3.3, the proposed method for incorporating corrosion pits into the FE model involves selecting a representative S-N curve parameters from DNV for an air environment. This approach diverges from diminishing the chain's fatigue life through the application of parameters (a) and (m) from the S-N curve in seawater, as provided by DNV guidelines. Instead, corrosion pits are directly integrated into the FE model, allowing for precise stress analysis within the pits themselves. Subsequently, fatigue life evaluation employs the parameters of the S-N curve associated with the air environment,

operating under the assumption that the corrosion does not affect the slope of the S-N curve.

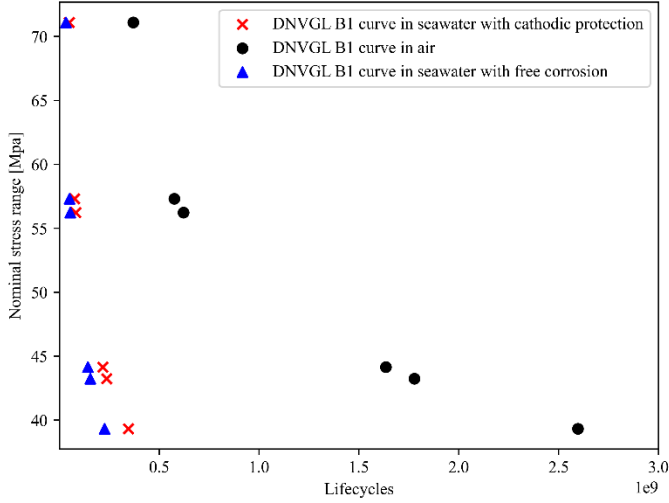
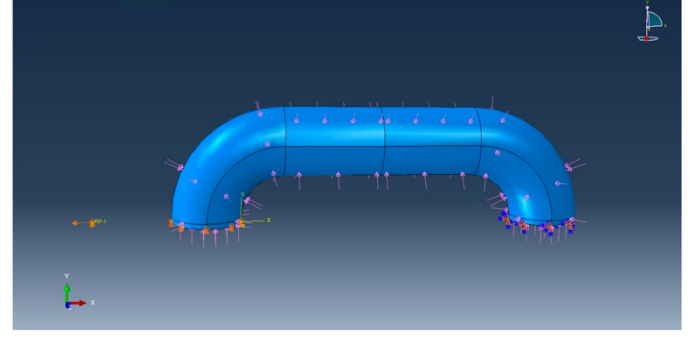


FIGURE 7: S-N CURVES ACCORDING TO DNV [18]

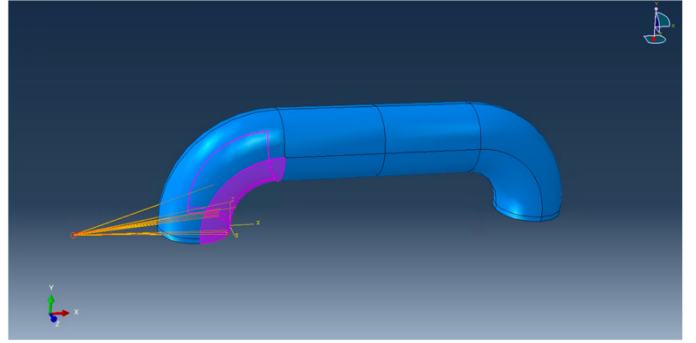
3.3 Finite Element Analysis

In the FE analysis conducted in ABAQUS/Standard, a half-chain model of the mooring line was examined (see Figure 8) following the geometrical specifications outlined in Chapter 2.1. This model was constructed based on the nominal diameter of the mooring line chain, which is 160 mm. The material properties attributed to this chain, made of R4 grade mooring steel, were incorporated based on the specifications provided by T. Lassen et al. [22]. The utilization of plastic deformation properties ensured that the model accurately reflected the physical and mechanical characteristics of the actual mooring line, allowing for a more precise and reliable analysis of its behavior under various conditions.

For analysis, only half of the mooring chain is modelled for computational efficiency. One side of the half-chain model, opposite to where the forces were to be introduced, was completely fixed in all directions, as shown in Figure 8A. Instead, the other side of the chain designated for force application, a boundary condition implementing a symmetry constraint ($U_2=U_{R1}=U_{R3}=0$) was employed. This symmetry constraint effectively represents the unmodeled half of the mooring chain, ensuring that the behavior of the entire chain is accurately reflected in the half-chain model's response to applied loads.



A)



B)

FIGURE 8: FE MODEL BOUNDARY CONDITIONS AND LOADS

To model the load distribution on the mooring chain while linked with the other chain segment, a reference point (RP-1) was used and coupled to the inner part of the model, as shown in Figure 8B. The coupling allows a load to be applied to a reference diameter on the inner bend of the chain, similar to the force of another mooring chain segment acting on the model. The DEL, applied as a load, was defined along the x- and y- axes at the RP-1. This strategic placement and orientation of the load effectively simulate the out-of-plane bending stresses that the mooring chain would encounter in marine conditions. Moreover, hydrostatic pressure was applied to the entire surface of the mooring chain, as shown in Figure 8A, considering the water depth and the density of seawater of 1025kg/m^3 .

A mesh sensitivity analysis was performed to optimize the balance between result accuracy and computational efficiency for the FE model. Increasing the number of elements leads to a more precise stress distribution but at the cost of longer computational times. The findings of this study are depicted in Figure 9, which indicates that a mesh size of a hexagon-shaped 7 mm elements, yielding to 70518 elements, was chosen as the most suitable for our analysis.

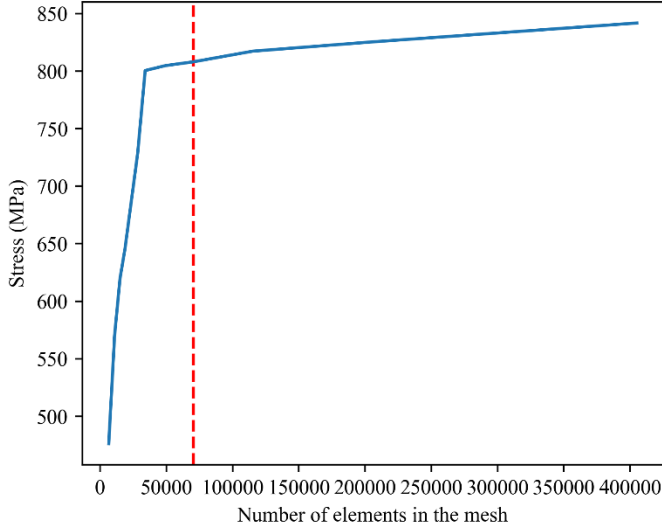


FIGURE 9: MESH SENSITIVITY ANALYSIS

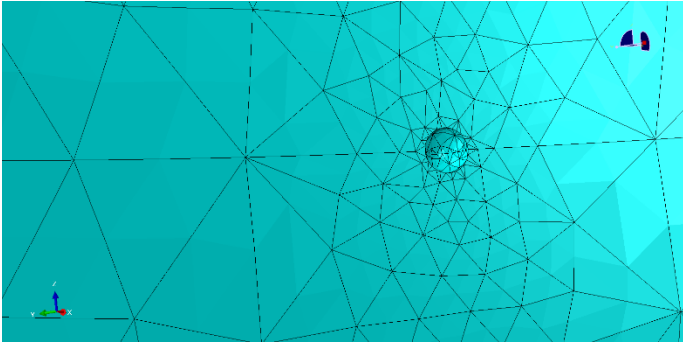


FIGURE 10: CORROSION PIT IN THE BEND SECTION OF THE MOORING CHAIN MODEL

The implementation of corrosion effects within the chain model was methodically applied to three critical locations identified as most susceptible to degradation: the crown, the bend section, and the weld section. These sections are recognized as inherent weak points within a mooring chain, as highlighted by Mendoza et al. [12]. Corrosion damage was simulated by incorporating pits as holes in the model, with their depths defined by the corrosion model detailed in Section 3.2. The mesh around each hole was refined to capture the geometry of the corrosion pits accurately, allowing for a precise representation of the pit's curvature as illustrated in Figure 10.

4. RESULTS AND DISCUSSION

This chapter presents the outcomes of the proposed methodology for the fatigue and the corrosion resistance evaluation of the mooring chain.

To comprehensively address fatigue and corrosion, the initial step involved calculating the loads, where the simulations of probable load scenarios were conducted. These simulations encompassed 12 distinct load cases for each critical point within the mooring chain, reported in Section 3.3. In Figure 11, ‘Seed

1’, ‘Seed 2’, and ‘Seed 3’ denote the generation of different load cases by generating random phases. Subsequently, from these simulations, the most critical load case for each location was identified. This case was characterized by a DEL that best represented the potential wear and tear at each point. These selected load cases were then utilized in further FE analysis.

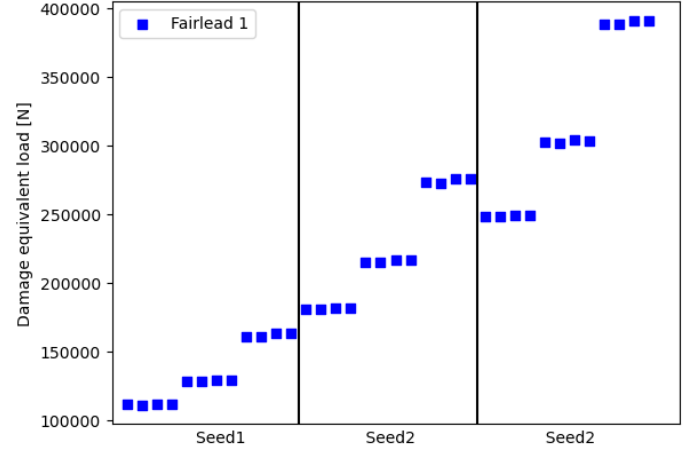


FIGURE 11: LOAD CASE PRESENTATION IN RELATION TO RANDOM PHASE GENERATION

The analysis of the mooring chain begins with an evaluation in an air environment before considering the impacts of corrosion. Initially, the chain's structural integrity is assessed without the influence of corrosion pits, focusing on fatigue life predictions. Table 4 presents the lifecycle estimates derived from the free corrosion curve, reflecting fatigue life based solely on the applied loads without the inclusion of corrosion effects.

The load angle and DEL are determined using OpenFAST, while the resulting stresses are obtained from FE analysis.

The evaluation of the fatigue life for chain segments adheres to the criteria and parameters set by DNV [18], specifically referencing material B1. Although the material B1 curve does not directly correlate with R4 steel chain, the comparative [9] use is justified based on the similarity of their properties. Furthermore, Section 6.2.2 of DNVGL-OS-E301 [23] suggests using the B1 curve parameters from [18] to assess fatigue life. This assessment incorporates stress concentration factors derived from the FE method as described in the presented study.

In an air environment—absent of corrosive factors—the bottom section emerges as the most critical area compared to the other sections, with a shorter fatigue life. This occurred due to the higher load levels with the higher resulting stresses at the bottom section of the mooring line.

The implementation of the corrosion pits in the FE model, as described in Section 3.3, illustrates an increase in stress levels, particularly in the crown and bend locations, as indicated in Figure 12. The higher stresses resulted in decreased fatigue life in all the sections, as reported in Table 5.

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TABLE 4: FATIGUE LIFE ASSESSMENT IN THE AIR

	Angle of load [rad]	DEL [kN]	$\Delta\sigma$ [MPa]	N_{air}
Top	0.77	180.4	39.37	2587.3 e+6
Middle	0.58	181.6	43.24	1778.1 e+6
Bottom	0.14	190.1	57.32	575.5 e+6

TABLE 5: FATIGUE LIFE ASSESSMENT IN SEAWATER (WITH CORROSION PITS)

	Angle of load [rad]	DEL [kN]	$\Delta\sigma$ [MPa]	N_{pit}
Top	0.77	180.4	89.63	12.8 e+6
Middle	0.58	181.6	98.28	8.8 e+6
Bottom	0.14	190.1	111.78	5.3 e+6

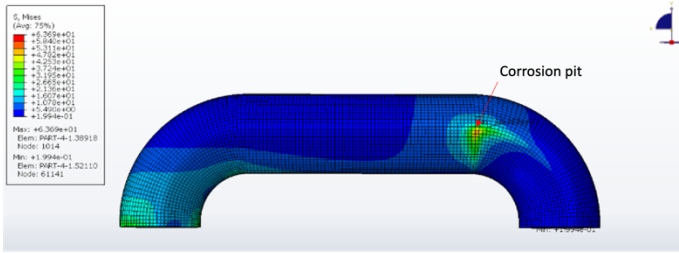


FIGURE 12: CORROSION PIT INDUCED STRESSES

A scatter plot including the different S-N results based on the air curve is presented in Figure 13 to compare the results of the different locations of corrosion pits. The presented results show a high reduction in fatigue life. As the corrosion pit is implemented in the bend where the coupling diameter is located, the force acts directly onto the corrosion pit, which could lead to high uncertainties. Implementing the pit in the crown section, as presented in Table 5, gives the highest lifecycle decrease.

Figure 14 displays the load cases that were analyzed for the effect of corrosion. It is possible to observe that the lifespan is significantly reduced in all cases due to the impact of the pit.

The load cases described in Table 3 were studied at three separate locations: top, middle, and bottom section of the catenary mooring line, which do not interact with the seabed or FOWT directly. Different pit depths were implemented at these three locations based on the model described in Section 3.2.

Table 6 summarizes the corrosion depth in different location of the mooring line and the resultant reduction of the fatigue life due to the development of the corrosion pits.

The analysis reveals a counterintuitive finding: despite the corrosion rate being highest at the top section of the mooring line, it is the bottom section that emerges as the most critical area in terms of fatigue life. This is attributed to the higher DEL observed at the bottom, as detailed in Section 3.1, when load conditions were examined. Should there be a shift in the direction of wind and current, coupled with a variation in load levels resulting in a reduced DEL at the bottom, the top section—with its greater corrosion rate—would likely become the predominant site for fatigue damage. This underscores the

complex interplay between environmental factors and structural integrity in the assessment of mooring line durability.

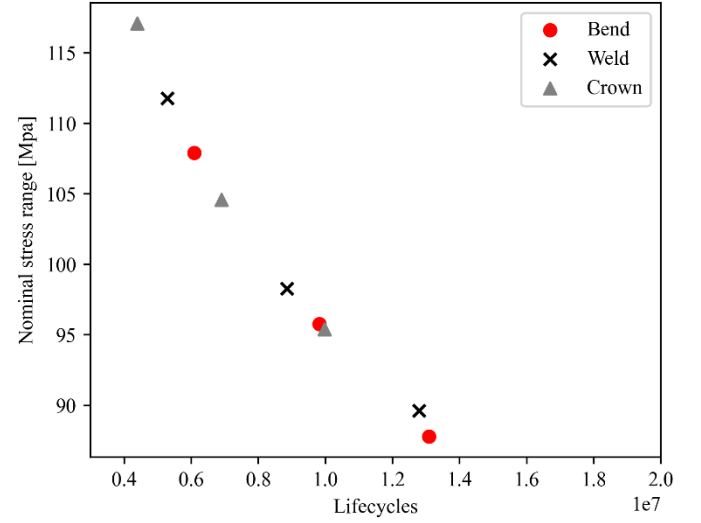


FIGURE 13: COMPARISON OF CORROSION IMPLEMENTATION LOCATIONS ON THE STRESS RESULT

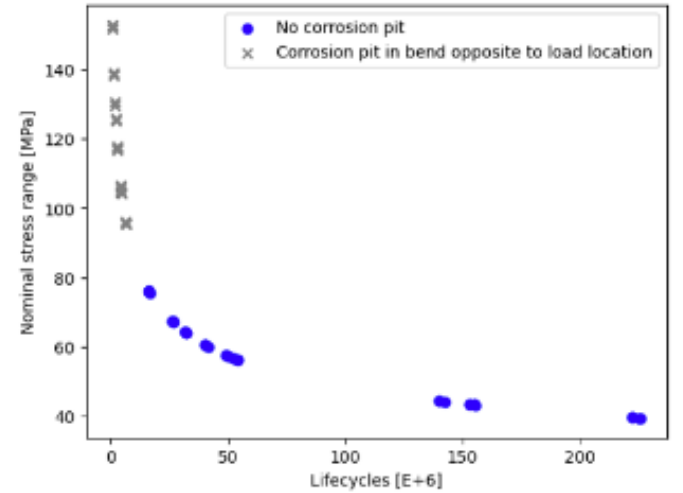


FIGURE 14: COMPARISON OF LIEFCYCLES OF PRE AND POST CORROSION PIT IMPLEMENTATION.

TABLE 6: FATIGUE LIFE ASSESSMENT AT DIFFERENT MOORING LINE LOCATIONS

Section	N_{air} (e+6)	C_D [mm]	N_{pit} (e+6)	Lifetime Reduction [%]
Top	225.32	0.120	9.96	95.6
Middle	154.85	0.114	6.90	95.5
Bottom	50.12	0.105	4.39	91.2

Figure 15 depicts the lowest life cycle count in the bottom section of the mooring chain when corrosion is not present, while the highest life expectancy is observed in the top section. However, following the implementation of corrosion, the highest reduction in life expectancy is observed in the top section. Although the lower section does not exhibit as significant a

reduction as the other sections, it still has the lowest life expectancy.

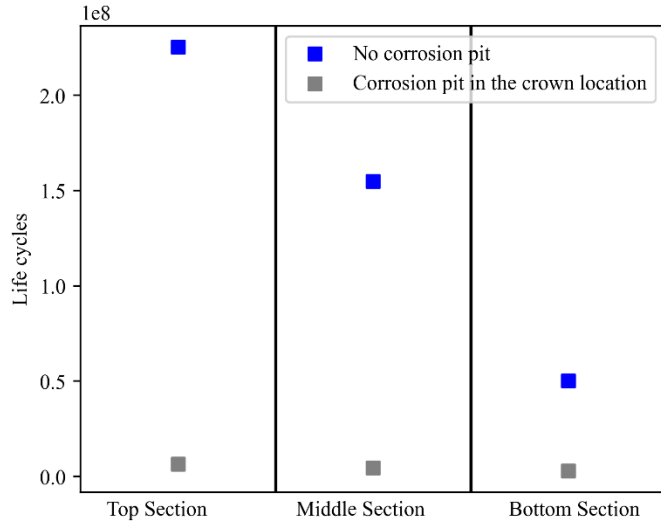


FIGURE 15: COMPARISON OF CORROSION IMPACT ON FATIGUE LIFE IN DIFFERENT MOORING LINE AREAS

The outcomes of the presented models indicate that the proposed methodology enables the identification of the combined effects of corrosion and fatigue along the mooring line, considering depth and immersion duration.

The findings correlate with the recent model developed based on Nevshupa and Fickian, where the corrosion pit is less than 1 mm within the 10-year immersion rate. Instead, after the immersion time of 10 years, the corrosion pits in the extracted chains grow at a higher rate than in the presented model. This disparity may arise due to disregard of microbial corrosion effect. As there is no data available for this location, the possibility of increased corrosion near the seabed and water surface cannot be excluded.

Since in the selected case study the mooring chain is located only 15 meters below sea level, there is no need to consider the chain's susceptibility to splash zone or microbial corrosion when determining its corrosion assumption. Thus, neglecting microbially induced corrosion only affects the corrosion rate near the seabed.

5. CONCLUSION

A methodology was developed to evaluate the cumulative degradation effects of corrosion and fatigue along the entire length of a mooring line. This approach involved conducting various load case simulations, both with and without the presence of corrosion pits. The results demonstrated that incorporating corrosion pits into the model significantly increases stress levels and reduces fatigue life. Furthermore, this methodology enabled the identification of weak points along the mooring chain through detailed fatigue and corrosion modeling.

REFERENCES

- [1] Tande, John Olav Gjøver, et al. "Floating offshore turbines." *Wiley Interdisciplinary Reviews: Energy and Environment* 4.3 (2015): 213-228.
- [2] Lloret, Josep, et al. "Unravelling the ecological impacts of large-scale offshore wind farms in the Mediterranean Sea." *Science of the Total Environment* 824 (2022): 153803.
- [3] Baniotopoulos, Charalampos. "Advances in Floating Wind Energy Converters." *Energies*, vol. 15, no. 15, 2021, p. 5658.
- [4] Fontaine, Emmanuel, et al. "Industry survey of past failures, pre-emptive replacements and reported degradations for mooring systems of floating production units." *Offshore technology conference*. OTC, 2014.
- [5] Ma, Kai-Tung, et al. *Mooring system engineering for offshore structures*. Gulf Professional Publishing, 2019.
- [6] Nevshupa, Roman, et al. "The effect of environmental variables on early corrosion of high-strength low-alloy mooring steel immersed in seawater." *Marine Structures* 60 (2018): 226-240.
- [7] Gabrielsen, Øystein, Turid Liengen, and Solfrid Molid. "Microbiologically influenced corrosion on seabed chain in the North Sea." *International Conference on Offshore Mechanics and Arctic Engineering*. Vol. 51227. American Society of Mechanical Engineers, 2018.
- [8] Du, Junfeng, et al. "Fatigue damage assessment of mooring lines under the effect of wave climate change and marine corrosion." *Ocean Engineering* 206 (2020): 107303.
- [9] Xue, Xutian, et al. "Mooring system fatigue analysis for a semi-submersible." *Ocean Engineering* 156 (2018): 550-563.
- [10] Angulo, Ángela, et al. "Mooring integrity management: Novel approaches towards in situ monitoring." *Structural Health Monitoring-Measurement Methods and Practical Applications* (2017): 87-108.
- [11] Orlikowski, Juliusz, et al. "Actual field corrosion rate of offshore structures in the Baltic Sea along depth profile from water surface to sea bed." *Ocean Engineering* 265 (2022): 112545.
- [12] Mendoza, Jorge, Per J. Haagen, and Jochen Köhler. "Analysis of fatigue test data of retrieved mooring chain links subject to pitting corrosion." *Marine Structures* 81 (2022): 103119.
- [13] He, Wentao, et al. "Numerical evaluation on fatigue crack growth and life predictions of an FPSO mooring system." *Ocean Engineering* 265 (2022): 112501.
- [14] Lone, Erling N., et al. "Fatigue assessment of mooring chain considering the effects of mean load and corrosion." *International Conference on Offshore Mechanics and Arctic Engineering*. Vol. 85123. American Society of Mechanical Engineers, 2021.
- [15] Vigara, Fernando et al. "D1.2 Design Basis." Technical report, 2020.
- [16] Zarandi, Ershad P., and Bjørn H. Skallerud. "Cyclic behavior and strain energy-based fatigue damage analysis

- of mooring chains high strength steel.” *Marine Structures* 70 (2020): 102703.
- [17] Boyer, Tim P., et al. “World ocean atlas.” (2018).
- [18] Veritas, Det Norske. “DNV-RP-C203: Fatigue design of offshore steel structures.” *DNV, Baerum, Norway* (2010).
- [19] Jonkman, Bonnie, and Jason Jonkman. “FAST v8. 16.00 a-bjj.” *National Renewable Energy Laboratory* 1355 (2016).
- [20] Cevasco, Debora, et al. “COREWIND D4.2 Floating Wind O&M Strategies Assessment-Condition Monitoring Strategies for Floating Wind O&M”.
- [21] Melchers, Robert E. “Predicting long-term corrosion of metal alloys in physical infrastructure.” *npj Materials Degradation* 3.1 (2019): 4.
- [22] Lassen, Tom, Eirik Storvoll, and Arild Bech. “Fatigue life prediction of mooring chains subjected to tension and out of plane bending.” *International Conference on Offshore Mechanics and Arctic Engineering*. Vol. 43413. 2009.
- [23] Veritas, Det Norske. “DNVGL-OS-E301: Position mooring.” *DNV, Baerum, Norway* (2015).